21 There are only red counters, yellow counters and blue counters in a bag.

Kevin takes at random a counter from the bag.
He puts the counter back in the bag.
Lethna takes at random a counter from the bag.
She puts the counter back in the bag.

The probability that both counters are red or that both counters are yellow is $\frac{13}{36}$

The probability that the first counter is red and the second counter is not red is $\frac{1}{4}$

Seb takes at random a counter from the bag.

Work out the probability that Seb takes a yellow counter.
You must show all your working.

Let r = P(red), y = P(yellow).

With replacement: $r^2 + y^2 = 13/36$

1 mark
Sum the two same-colour cases.

$r(1 - r) = 1/4$

1 mark
Red then not red.

$r^2 - r + 1/4 = 0$

$(r - ½)^2 = 0$, so $r = ½$

1 mark
Solve for r.

$y^2 = 13/36 - (½)^2$

$y^2 = 13/36 - 9/36 = 4/36 = 1/9$

1 mark
Substitute r to find y^2.

$y = \sqrt{(1/9)}$ (probability is positive)

$y = \frac{1}{3}$

1 mark
Take the positive root.

(Total for Question 21 is 5 marks)